A Project Proposal

For

BUS TICKET BOOKING MANAGEMENT SYSTEM



SUBMITTED BY

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Abstract

The Bus Ticket Booking Management System is a desktop-based application designed to simplify and automate the process of intercity bus ticket reservations and administrative operations. The system will be developed using Java Swing for the graphical user interface, JDBC for database connectivity, and MariaDB/MySQL through XAMPP for database management.

The application will provide two major user roles: Administrator and Customer. Customers will be able to register, log in, search for available bus trips, view seat availability through an interactive visual seat layout, select seats, choose payment methods, and generate digital boarding tickets. Administrators will be able to manage buses, routes, trip schedules, customer information, and booking records through a centralized dashboard.

The project will demonstrate important Database Management System concepts, including relational database normalization up to Third Normal Form (3NF), primary and foreign keys, SQL joins, subqueries, aggregate functions, and ACID transactions. JDBC PreparedStatement will be used to prevent SQL injection, while database locking mechanisms will help prevent duplicate seat reservations.

The main objective of this project is to develop a functional database-driven reservation system while demonstrating practical knowledge of Java programming, relational database design, transaction management, and software architecture.

Keywords: Bus Reservation, Java Swing, JDBC, MySQL, MariaDB, DBMS, ACID Transactions, Seat Booking.

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Chapter 1: INTRODUCTION

1.1 Problem Specification / Statement

In traditional bus ticket reservation systems, customers often need to visit physical ticket counters or communicate with booking agents to check bus schedules and seat availability. These processes can be time-consuming and may involve manual record management.

Manual reservation systems can create several problems, such as duplicate bookings, difficulties in maintaining accurate seat availability, delays in retrieving booking information, and challenges in generating financial reports.

To overcome these limitations, the proposed Bus Ticket Booking Management System will provide a centralized desktop-based solution for managing bus reservations and related administrative activities.

Customers will be able to search for trips, check available seats, select preferred seats, and generate digital tickets. Administrators will have the ability to manage buses, routes, schedules, and booking information.

The system will use a relational database to maintain consistency, accuracy, and reliability of information.

1.2 Objectives

The main objectives of this project are:

1. To develop a desktop-based bus ticket booking application using Java Swing.
2. To design and implement a normalized relational database using MySQL/MariaDB.
3. To provide separate login and dashboard facilities for administrators and customers.
4. To allow customers to search for available bus trips based on departure city, destination, and travel date.
5. To implement an interactive visual seat selection system.
6. To prevent duplicate seat reservations using database transaction management and locking mechanisms.
7. To provide bus, route, and trip management facilities for administrators.
8. To generate digital boarding tickets and allow customers to download ticket receipts.
9. To implement secure database operations using JDBC PreparedStatement.
10. To demonstrate practical DBMS concepts such as normalization, SQL joins, subqueries, and ACID transactions.

1.3 Motivation and Scope

The motivation behind this project is to develop a practical database-driven application that demonstrates how modern software systems can automate reservation management.

Through this project, we aim to apply our theoretical knowledge of Database Management Systems, Java programming, and software engineering to a real-world problem.

The scope of the system includes:

- User registration and authentication.
- Administrator and customer dashboards.
- Bus fleet management.
- Route management.
- Trip scheduling.
- Trip searching and filtering.
- Interactive bus seat selection.
- Booking and payment record management.
- Digital ticket generation.
- Booking history and cancellation.
- Administrative statistics and reports.

The initial version will be developed as a local desktop application. Online payment gateway integration, mobile applications, and real-time GPS tracking will remain outside the initial project scope.

1.4 Organization of Project Proposal

This proposal is organized into six chapters.

Chapter 1 introduces the project, including its problem statement, objectives, motivation, and scope.

Chapter 2 discusses existing reservation systems and the technical concepts related to the proposed application.

Chapter 3 presents the system analysis and design, including the technology stack, system architecture, database design, diagrams, algorithms, and requirements.

Chapter 4 explains the proposed implementation, including front-end design, back-end architecture, and system modules.

Chapter 5 describes the hardware and software requirements and provides an overview of the planned user interfaces.

Chapter 6 concludes the proposal by discussing the expected outcomes, limitations, and possible future improvements.

Chapter 2: BACKGROUND

2.1 Existing System Analysis

Currently, bus ticket reservations can generally be managed through manual counter-based systems or computerized reservation platforms.

Manual Reservation System

In a manual reservation system, customers visit ticket counters to collect information about available buses, schedules, and seats.

Advantages:

- Simple reservation process.
- Direct communication with booking staff.
- Requires limited technical infrastructure.

Disadvantages:

- Time-consuming booking process.
- Possibility of human errors.
- Difficulty maintaining updated seat availability.
- Manual record management.
- Difficulties in generating reports.

Computerized Reservation System

Computerized reservation systems store booking information in databases and allow users to manage reservations electronically.

Advantages:

- Faster trip searching.
- Better reservation management.
- Accurate seat availability information.
- Easier data retrieval.
- Improved reporting capabilities.

Disadvantages:

- Requires technical infrastructure.
- Requires proper database management.

- System maintenance and security are necessary.

The proposed system aims to provide the benefits of computerized reservation management through a simple Java-based desktop application.

2.2 Supporting Literatures

The proposed system is based on several important concepts of database management and software development.

Relational Database Management System (RDBMS):

An RDBMS organizes information into related tables. The proposed system will use MariaDB/MySQL to store users, buses, routes, trips, seats, bookings, and payment information.

Database Normalization:

Database normalization helps reduce data redundancy and maintain data consistency. The proposed database will follow Third Normal Form (3NF).

ACID Transactions:

ACID properties ensure reliable database transactions:

- Atomicity
- Consistency
- Isolation
- Durability

These properties will be applied during booking creation and bus registration.

JDBC:

Java Database Connectivity (JDBC) will establish communication between the Java application and the MySQL/MariaDB database.

Three-Tier Architecture:

The system will separate presentation, business logic, and database operations to improve maintainability and organization.

Chapter 3: SYSTEM ANALYSIS & DESIGN

3.1 Technology & Tools

Technology	**Purpose**
Java (JDK 17+)	Core programming language
Java Swing	Graphical User Interface
JDBC	Database connectivity
XAMPP	Local database environment
MariaDB/MySQL	Relational database
MySQL Connector/J 8.3.0	JDBC driver
phpMyAdmin	Database administration
Visual Studio Code	Code development
Windows 10/11	Target operating system

3.2 Model & Diagram

3.2.1 Development Model

The project will follow an Iterative and Incremental Development Model.

The development process will be divided into several stages:

1. Requirement Analysis
2. Database Design
3. GUI Design
4. Authentication Development
5. Admin Module Development
6. Customer Module Development
7. Booking and Transaction Implementation
8. Testing and Debugging
9. Final Integration and Documentation

Each module will be developed and tested independently before integrating it into the complete application.

3.2.2 System Architecture

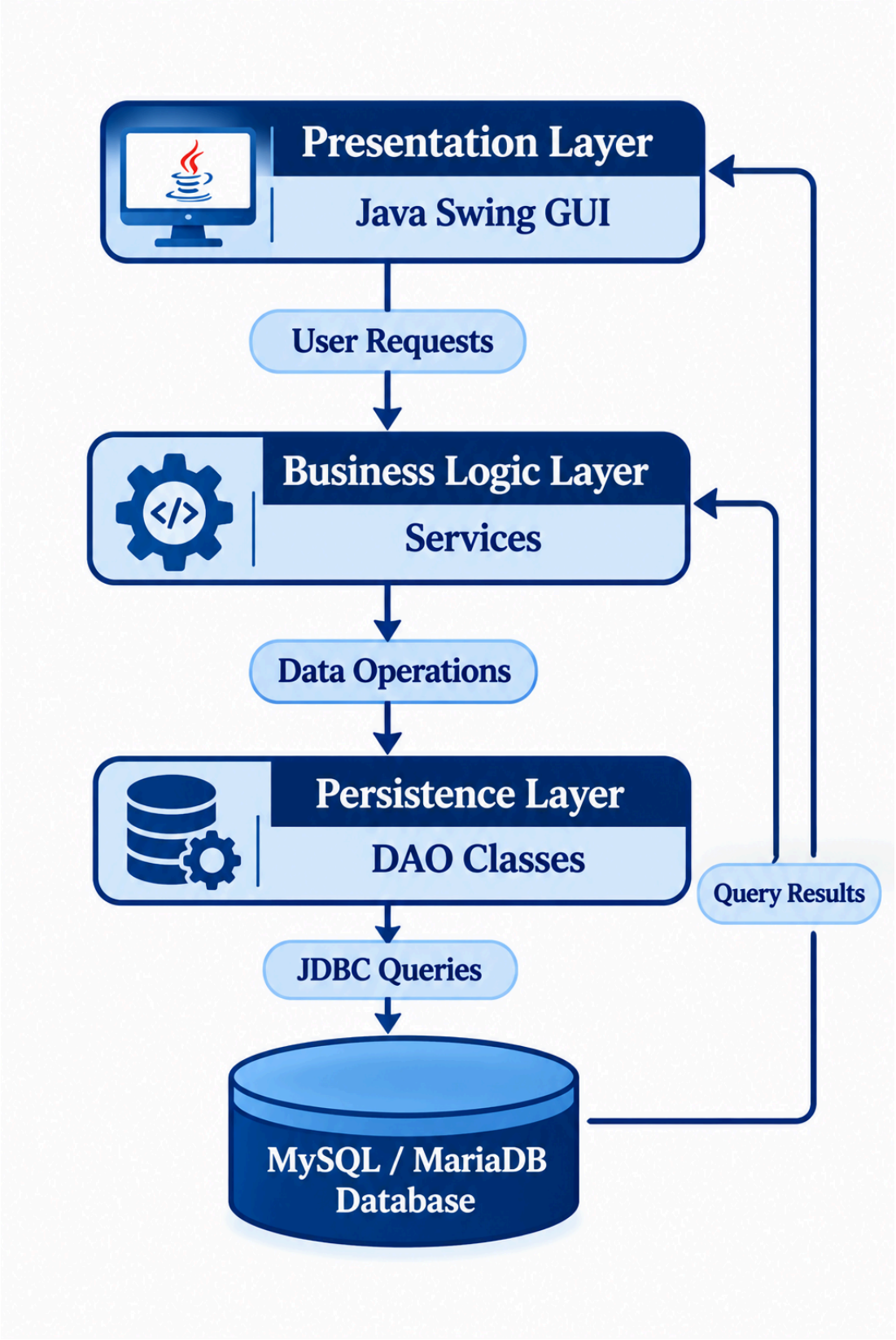


Figure 1: Three-Tier System Architecture

3.2.3 Use Case Overview

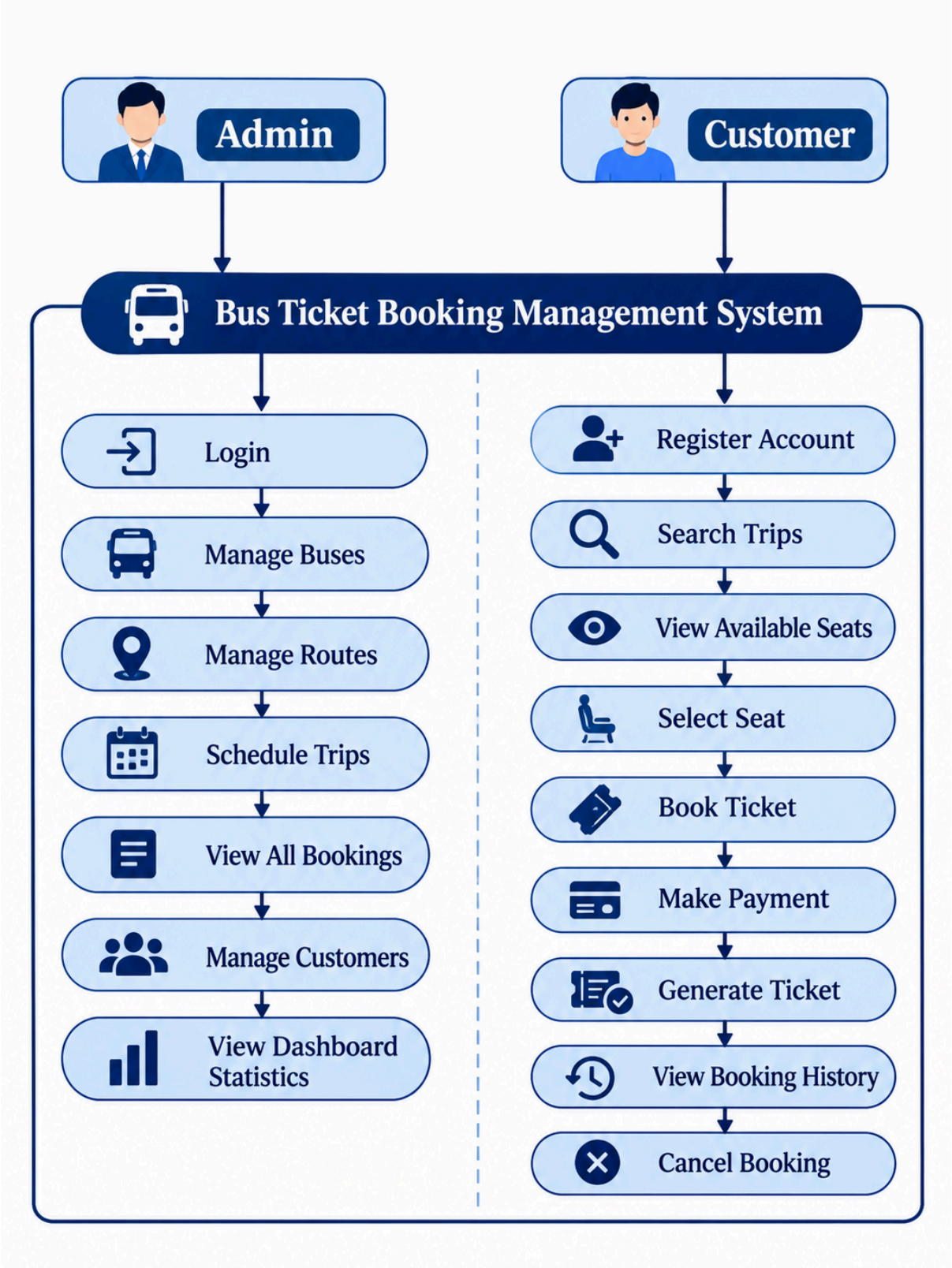


Figure 2: Use Case Diagram

3.2.4 ER Diagram

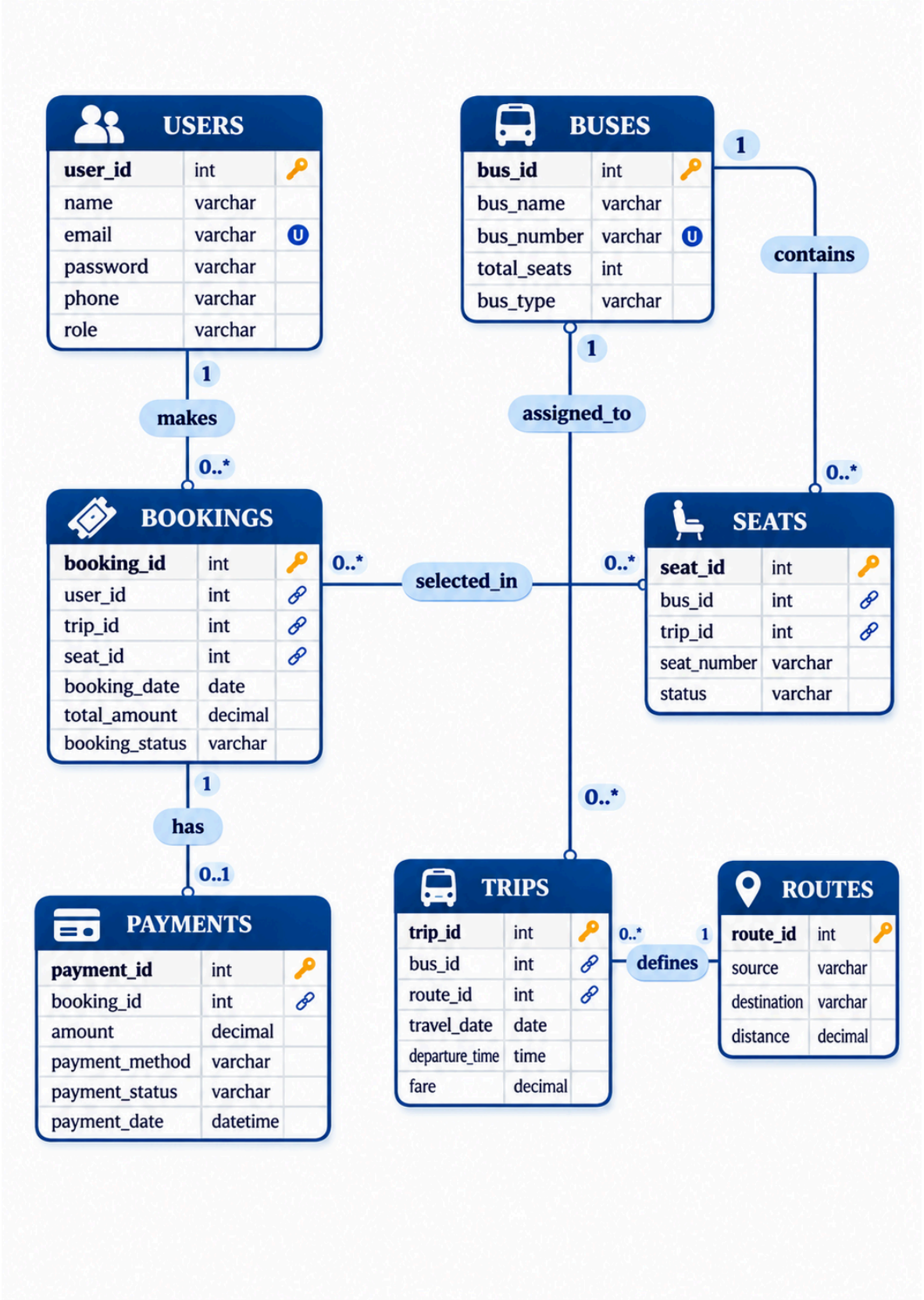


Figure 3: Entity Relationship (ER) Diagram

3.2.5 Data Flow Diagram

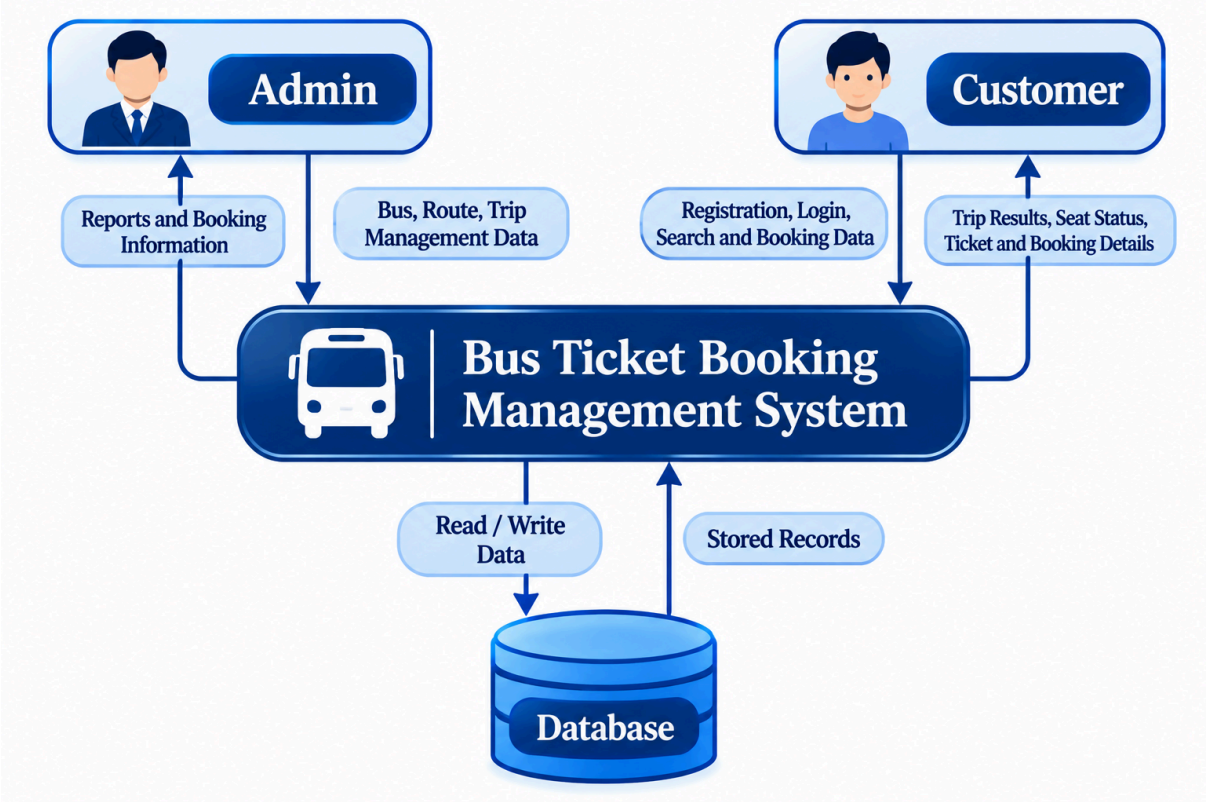


Figure 4 : Data Flow Diagram (DFD)

3.2.6 Database Schema

Table Name	Main Attributes
Users	user_id (PK), name, email, password_hash, role, created_at
Buses	bus_id (PK), bus_name, bus_number, bus_type, total_seats
Routes	route_id (PK), departure_city, destination_city, distance, duration
Trips	trip_id (PK), bus_id (FK), route_id (FK), travel_date, departure_time, fare, status
Seats	seat_id (PK), bus_id (FK), seat_number, seat_type
Bookings	booking_id (PK), user_id (FK), trip_id (FK), seat_id (FK), booking_time, fare, status
Payments	payment_id (PK), booking_id (FK), amount, payment_method, payment_status, payment_time

3.2.7 Booking Algorithm / Flowchart

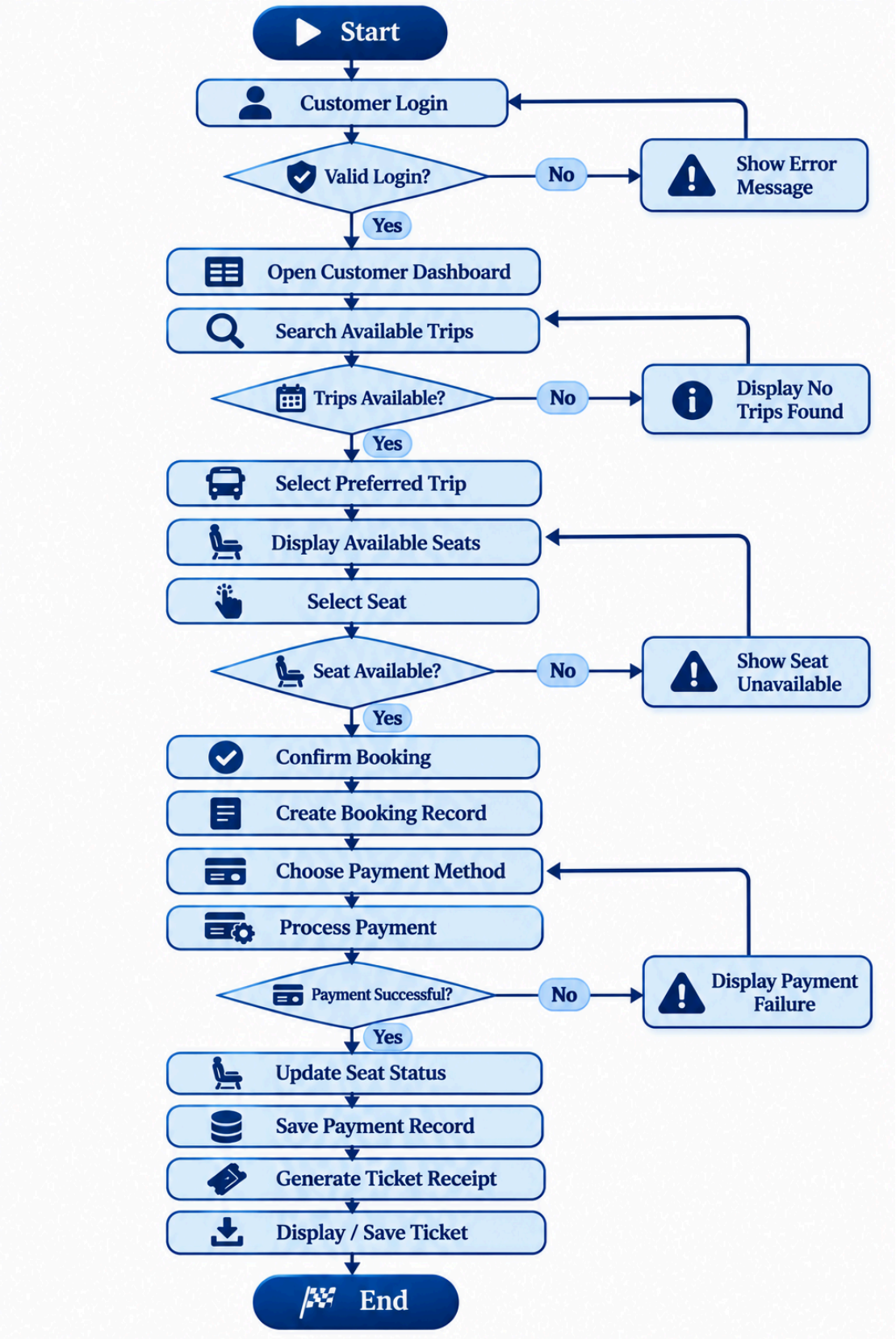


Figure 5: Booking Process Flowchart

3.3 Features and Requirements

3.3.1 Features

- Role-based authentication.
- Customer self-registration.
- Admin dashboard.
- Bus fleet management.
- Route management.
- Trip scheduling.
- Trip search.
- Interactive bus seat selection.
- Color-coded seat availability.
- Payment method selection.
- Digital boarding pass.
- Ticket download.
- Booking history.
- Booking cancellation.
- Administrative reports.

3.3.2 Functional Requirements

ID	Requirement
FR-01	The system shall authenticate administrators and customers.
FR-02	Customers shall be able to register accounts.
FR-03	Administrators shall manage bus information.
FR-04	Administrators shall manage routes and trips.
FR-05	Customers shall search for available trips.
FR-06	The system shall display seat availability.
FR-07	Customers shall be able to book available seats.
FR-08	The system shall prevent duplicate seat reservations.
FR-09	The system shall generate digital tickets.
FR-10	Customers shall view their booking history.
FR-11	Customers shall be able to request booking cancellation.
FR-12	Administrators shall view system statistics.

3.3.3 Non-functional Requirements

- Security: Protect user information and prevent SQL injection.
- Reliability: Maintain database consistency using transactions.
- Performance: Retrieve trip and booking information efficiently.
- Usability: Provide a simple and interactive graphical interface.
- Maintainability: Follow a three-tier software architecture.
- Data Integrity: Maintain relationships and constraints between database tables.

Chapter 4: PROPOSED IMPLEMENTATION

4.1 Interface / Front-End Design

The front end will be developed using Java Swing and AWT.

The application will contain a graphical login screen, registration screen, administrator dashboard, customer dashboard, seat selection interface, ticket display, and booking history interface.

The visual seat selection interface will represent a standard 40-seat intercity bus.

The seat layout will follow a 2×2 arrangement with a central aisle.

The seat colors will represent:

- Green: Available Seat
- Light blue: Selected Seat
- Cyan: Sold Seat

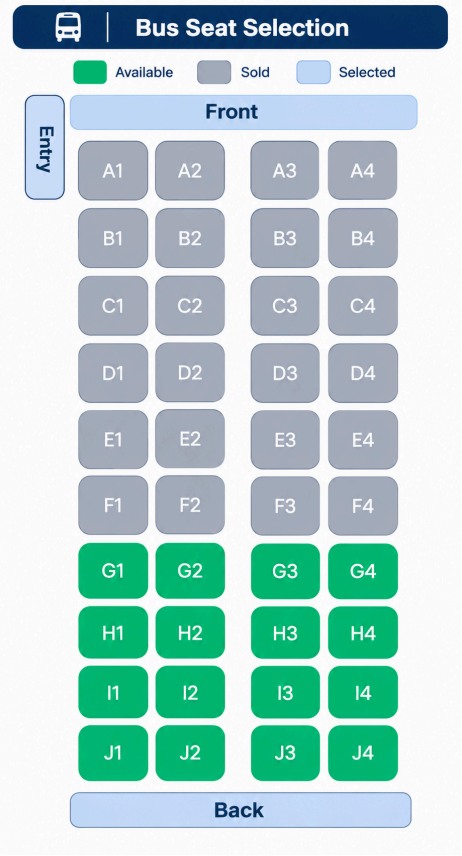


Figure 6: Bus Seat Selection Layout

4.2 Back-End Design

The back end will be developed using Java and JDBC.

The application will communicate with the MariaDB/MySQL database through the MySQL Connector/J driver.

The DAO classes will handle database operations, while service classes will manage business logic.

PreparedStatement will be used for parameterized SQL queries.

Database transactions will be implemented using:

- `conn.setAutoCommit(false)`
- `conn.commit()`
- `conn.rollback()`

These operations will ensure that booking and payment records are processed consistently.

4.3 Proposed Modules

1. Authentication & User Management

Handles administrator login, customer login, customer registration, and role-based navigation.

2. Admin Dashboard

Displays total bookings, total revenue, fleet count, active routes, and registered customers.

3. Bus Fleet Management

Allows administrators to add, update, and delete buses. Adding a new bus automatically generates 40 physical seat records.

4. Route Management

Allows administrators to manage departure cities, destinations, distances, and journey durations.

5. Trip Scheduling

Allows administrators to assign buses and routes to specific travel dates and times.

6. Trip Search

Allows customers to search for trips using departure city, destination, and travel date.

7. Seat Selection

Displays available and booked seats through an interactive visual interface.

8. Booking & Payment Management

Processes reservations and stores payment information using database transactions.

9. Digital Ticket Generation

Generates digital boarding tickets and allows customers to save them as text files.

10. Booking History & Cancellation

Allows customers to view previous reservations and request ticket cancellation.

Chapter 5: PROPOSED USER MANUAL

5.1 System Requirements

5.1.1 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB
Storage	1 GB available space
Display	1366 × 768 resolution

5.1.2 Software Requirements

Software	Requirement
Operating System	Windows 10/11
Java	JDK 17 or later
Database Server	XAMPP
Database	MariaDB/MySQL
JDBC Driver	MySQL Connector/J
IDE	Visual Studio Code
Database Interface	phpMyAdmin

5.2 Planned User Interfaces

5.2.1 Login and Registration

The login interface will allow administrators and customers to access the system using their registered email and password.

New customers will be able to create accounts through the registration interface.

The login screen will also provide demo credential buttons and a database connection testing option.

5.2.2 Admin Dashboard

The administrator dashboard will provide access to:

- Overview and Statistics
- Bus Fleet Management
- Route Management
- Trip Scheduling
- All Bookings
- Registered Customers

5.2.3 Customer Booking Portal

The customer dashboard will allow users to:

- Search available trips.
- View bus schedules.
- Select available seats.
- Choose payment methods.
- Confirm bookings.
- View and download digital tickets.
- View booking history.
- Cancel reservations.

5.2.4 Demo Credentials

Role	Name	Email	Password
Admin	Tanvir Admin	admin@busticket.com	admin123
Customer	Rahim Ahmed	rahim@gmail.com	rahim123
Customer	Fatima Begum	fatima@gmail.com	fatima123
Customer	Karim Khan	karim@gmail.com	karim123

Chapter 6: CONCLUSION

6.1 Expected Outcome

The expected outcome of this project is a fully functional Java Swing-based Bus Ticket Booking Management System integrated with a relational database.

The application will provide separate interfaces for administrators and customers, interactive seat booking, trip searching, bus management, route management, digital ticket generation, and booking history.

From an academic perspective, the project will demonstrate practical implementation of database normalization, relational database relationships, SQL queries, JDBC connectivity, ACID transactions, concurrency control, and three-tier software architecture.

The project will also help us gain practical experience in developing database-driven desktop applications using Java and MySQL/MariaDB.

6.2 Project Limitations

The initial version of the system will have some limitations:

1. The application will operate as a local desktop application.
2. An active local database connection will be required.
3. Real online payment gateway integration will not be available.
4. Live GPS-based bus tracking will not be implemented.
5. Email and SMS notification services will not be included.
6. Bus schedules and route information will be maintained by administrators.
7. Automatic financial refunds will not be processed through real banking services.

6.3 Future Works

In the future, the system can be improved by introducing:

1. Web-based bus ticket booking facilities.
2. Android and iOS mobile applications.
3. Online payment gateway integration.
4. QR-code-based digital tickets.
5. Email and SMS notifications.
6. Live GPS bus tracking.
7. Multiple bus operator management.

8. Advanced booking analytics and reporting.
9. Online database hosting and multi-branch support.

References

1. Oracle. Java Documentation. https://docs.oracle.com/en/java/
2. Oracle. JDBC Basics. https://docs.oracle.com/javase/tutorial/jdbc/
3. MySQL. MySQL Reference Manual. https://dev.mysql.com/doc/refman/8.0/en/
4. MySQL. MySQL Connector/J Developer Guide. https://dev.mysql.com/doc/connector-j/en/
5. Apache Friends. XAMPP. https://www.apachefriends.org/
6. Elmasri, Ramez, and Shamkant B. Navathe. Fundamentals of Database Systems. Pearson.

Appendix A: Project Directory Structure

